DESCENT DATA FOR QUASI-AFFINE MORPHISMS OVER AN FPQC COVERING ARE EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Effectivity of descent data for morphisms whose type descends, over an fpqc covering. Same chapter, Lemma lemma-descending-types-morphisms, invoked at proof lines 9465-9468. Reduces the general fpqc covering to the case of a single flat surjective morphism of affines.
- The characterisation of quasi-affine morphisms by a quasi-compact open immersion into the relative spectrum of the pushforward. Morphisms, Lemmas morphisms-lemma-characterize-quasi-affine and morphisms-lemma-base-change-quasi-affine, invoked at proof lines 9461-9464 and 9475-9480. Produces the open immersion V -> W and the base-change stability used for the reduction.
- Flat base change for the cohomology of quasi-coherent sheaves. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology, invoked at proof lines 9489-9490. Identifies the pullbacks pr_0^* pi_* O_V and pr_1^* pi_* O_V with the pushforwards of the two base changes of V, so that phi induces phi^sharp.
- Effectivity of descent for affine morphisms, and affineness of the descended scheme. Same chapter, Lemmas lemma-affine and lemma-descending-property-affine, invoked at proof lines 9506-9511. Produces the scheme U' descending W and records that U' is affine.
- The quotient topology of an fpqc covering. Morphisms, Lemma morphisms-lemma-fpqc-quotient-topology, invoked at proof lines 9528-9529. Shows that the descended subset U of U' is open, which finishes the proof.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0245: sources/stacks-descent/pinned-original-source.tex, source lines 9365–9365. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02L5: sources/stacks-descent/pinned-original-source.tex, source lines 6143–6143. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02W3: sources/stacks-descent/pinned-original-source.tex, source lines 9245–9245. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01SO: sources/stacks-chow/tags.txt, source lines 2345–2345. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01SM: sources/stacks-chow/tags.txt, source lines 2343–2343. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02JY: sources/stacks-chow/tags.txt, source lines 3353–3353. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 022B: sources/stacks-chow/tags.txt, source lines 2700–2700. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\{X_i \to S\}_{i \in I}$ be an fpqc covering, see Topologies, Definition 022B. Let $(V_i/X_i, \varphi_{ij})$ be a descent datum relative to $\{X_i \to S\}$. If each morphism $V_i \to X_i$ is quasi-affine, then the descent datum is effective.*

Proof. Being quasi-affine is a property of morphisms of schemes which is preserved under any base change, see Morphisms, Lemmas 01SM and 01SO. Hence Lemma 02W3 applies and it suffices to prove the statement of the lemma in case the fpqc-covering is given by a single $\{X \to S\}$ flat surjective morphism of affines. Say $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(R)$ so that $R \to A$ is a faithfully flat ring map. Let (V, φ) be a descent datum relative to X over S and assume that $\pi : V \to X$ is quasi-affine.

According to Morphisms, Lemma 01SM this means that

$$V \longrightarrow \underline{\mathrm{Spec}}_X(\pi_*\mathcal{O}_V) = W$$

is a quasi-compact open immersion of schemes over X. The projections $\mathrm{pr}_i : X \times_S X \to X$ are flat and hence we have

$$\mathrm{pr}_0^*\pi_*\mathcal{O}_V = (\pi \times \mathrm{id}_X)_*\mathcal{O}_{V \times_S X}, \quad \mathrm{pr}_1^*\pi_*\mathcal{O}_V = (\mathrm{id}_X \times \pi)_*\mathcal{O}_{X \times_S V}$$

by flat base change (Cohomology of Schemes, Lemma 02KH). Thus the isomorphism $\varphi : V \times_S X \to X \times_S V$ (which is an isomorphism over $X \times_S X$) induces an isomorphism of quasi-coherent sheaves of algebras

$$\varphi^\sharp : \mathrm{pr}_0^*\pi_*\mathcal{O}_V \longrightarrow \mathrm{pr}_1^*\pi_*\mathcal{O}_V$$

on $X \times_S X$. The cocycle condition for φ implies the cocycle condition for $\varphi^\sharp$. Another way to say this is that it produces a descent datum φ' on the affine scheme W relative to X over S, which moreover has the property that the morphism $V \to W$ is a morphism of descent data. Hence by Lemma 0245 (or by effectivity of descent for quasi-coherent algebras) we obtain a scheme $U' \to S$ with an isomorphism $(W, \varphi') \cong (X \times_S U', can)$ of descent data. We note in passing that U' is affine by Lemma 02L5.

And now we can think of V as a (quasi-compact) open $V \subset X \times_S U'$ with the property that it is stable under the descent datum

$$can : X \times_S U' \times_S X \to X \times_S X \times_S U', (x_0, u', x_1) \mapsto (x_0, x_1, u').$$

In other words $(x_0, u') \in V \Rightarrow (x_1, u') \in V$ for any x_0, x_1, u' mapping to the same point of S. Because $X \to S$ is surjective we immediately find that V is the inverse image of a subset $U \subset U'$ under the morphism $X \times_S U' \to U'$. Because $X \to S$ is quasi-compact, flat and surjective also $X \times_S U' \to U'$ is quasi-compact flat and surjective. Hence by Morphisms, Lemma 02JY this subset $U \subset U'$ is open and we win. $\square$